

THE EMERGENT BELT

涌现之带

Jing-Zhang Hyper Line

Centennial Jing-Zhang AI Belt · open-call submission

KUN-SAL Spatial Quant Lab · Xing Xinhua



$\lambda_c=0.42$

connection

$H=1.556$

health

$D=1.746$

fractal

25,476

POIs

+15.9%

job reach

44亿

investment

Conceptual: provisional boundary dashed only; statutory indicators pending

01

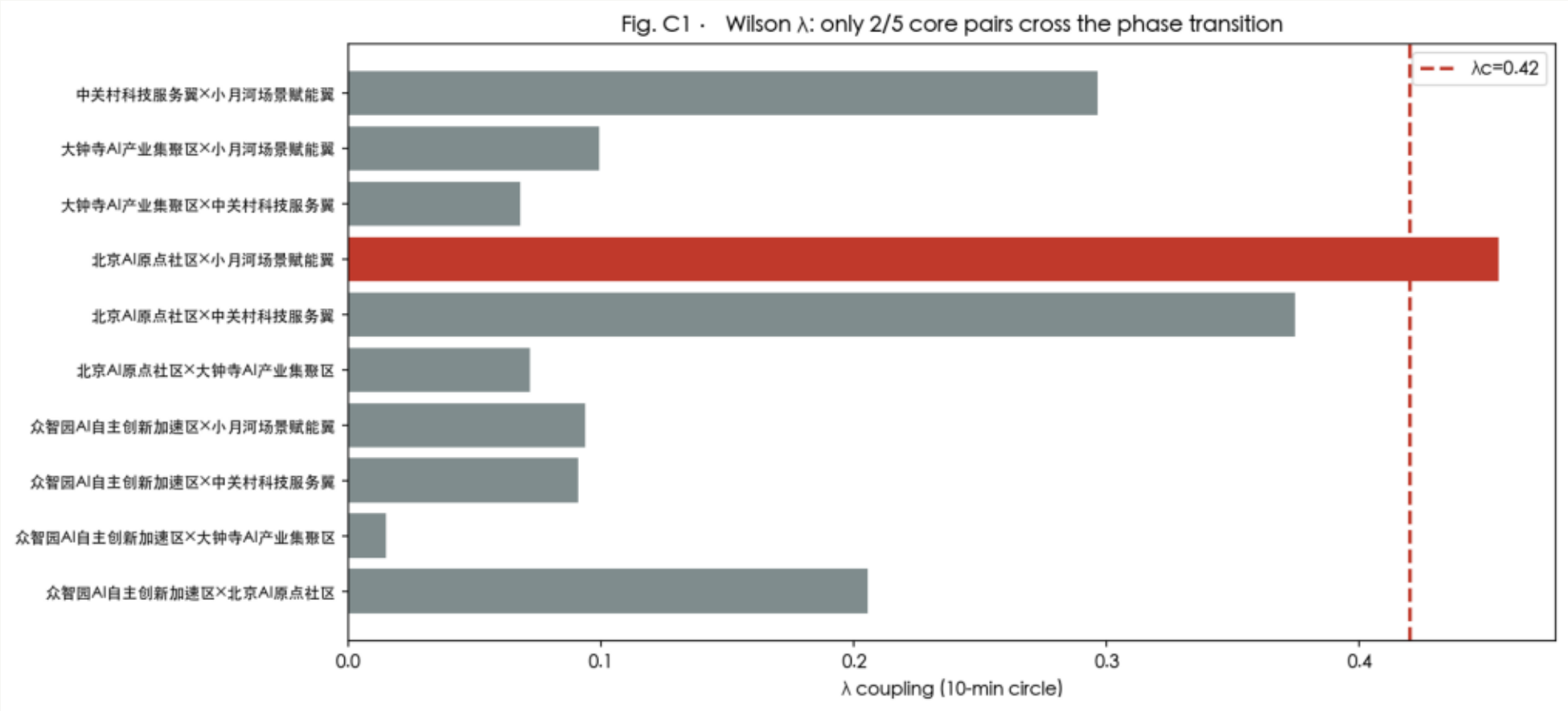
Diagnosis: what blocks this belt

Five scientists' consultation · Space syntax · POI dynamics · Basis & boundary correction

① Spatial interaction λ: stuck subcritical

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- Only one of ten core pairs crosses the line (Origin × Xiaoyuehe 0.455)
- Zhongzhiyuan×Dazhongsi λ=0.033: adjacent on the map, unreachable in ten minutes
- Not without vitality — vitality chopped apart by seven roads
- The leverage point is the seven gaps: no even spreading



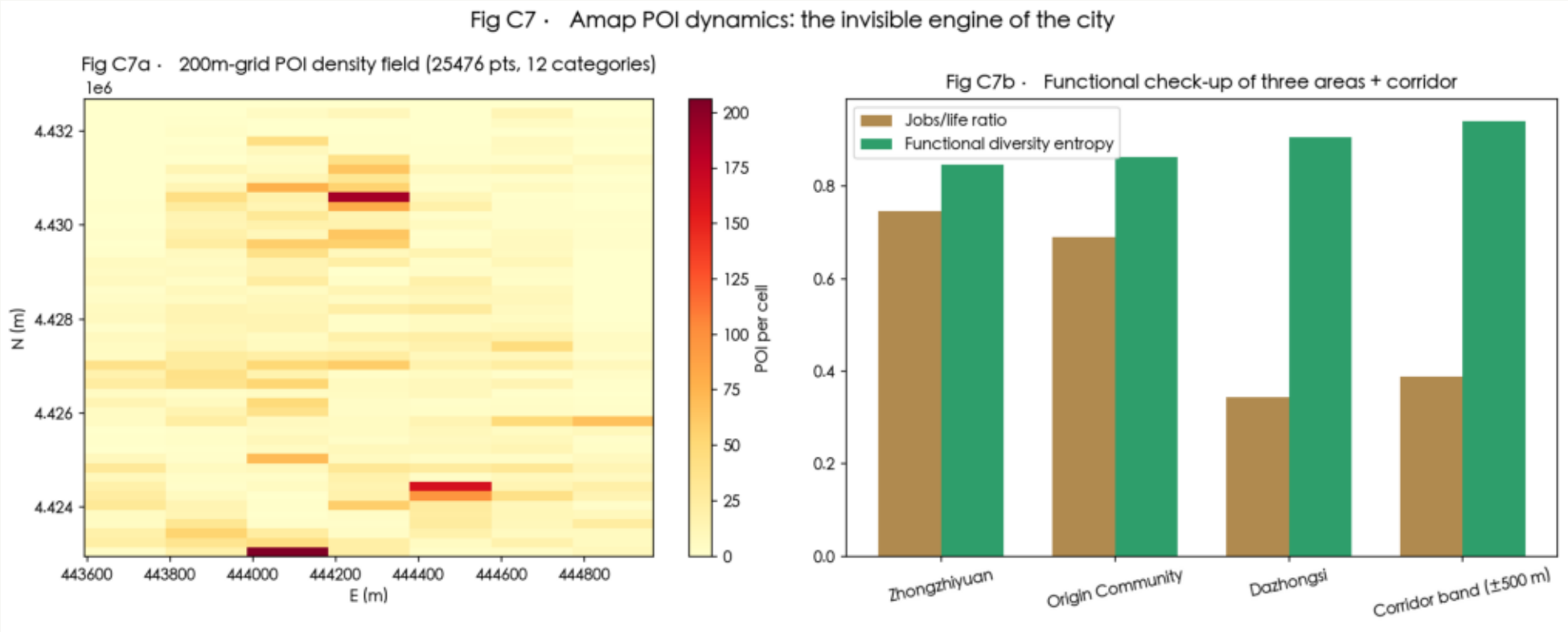
Basis: $\lambda = \exp(-d/2500m)$ proxy; recalculation on official data



② POI check-up: demand recognized by 25,476 points

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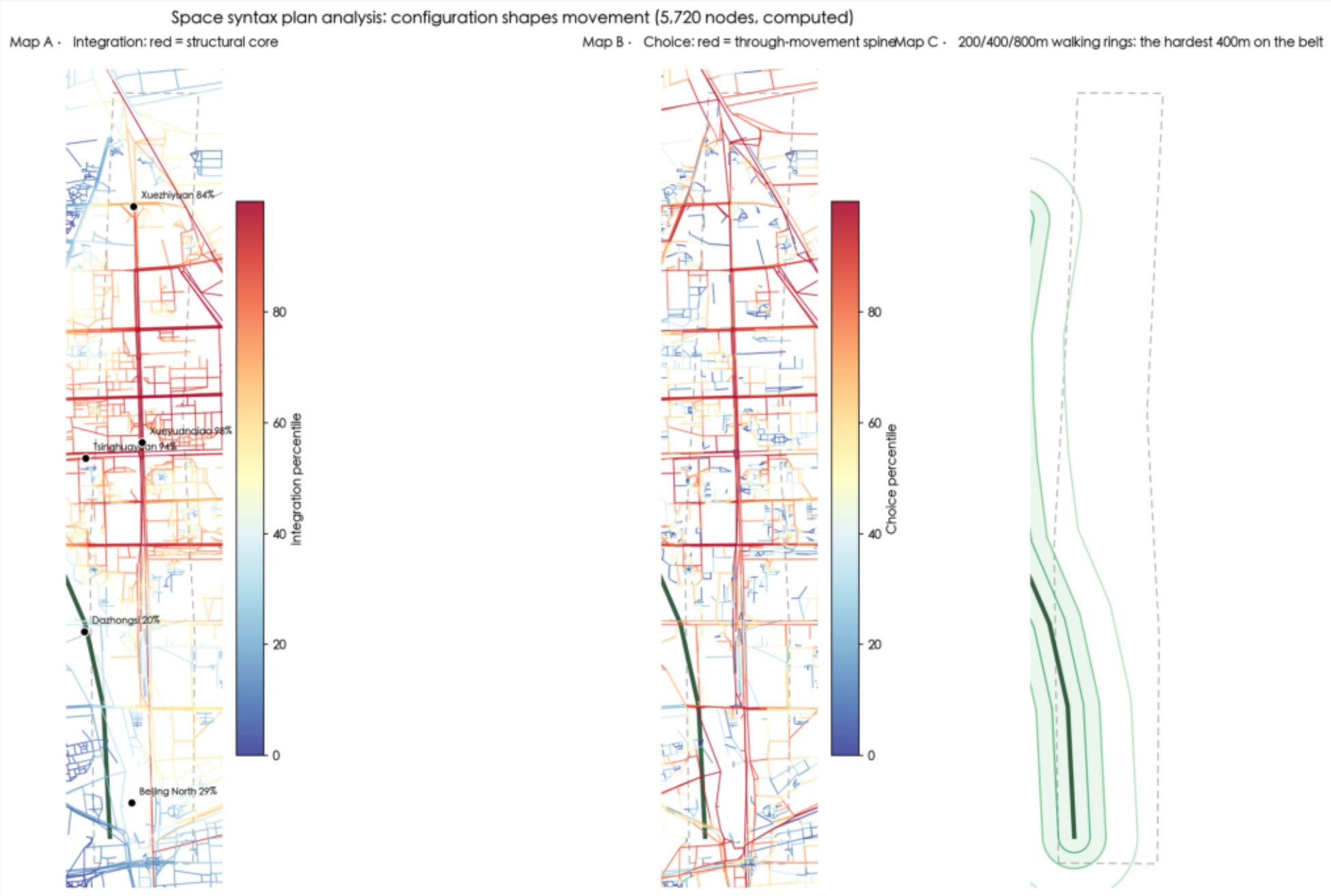
- 25,476 POIs across the belt (2026-08-15 snapshot)
- Life services over-supplied; innovation content under-supplied
- Finance –63.0% / research –45.9% / Origin medical 2.88/km², the lowest
- Dazhongsi 511/km², the densest; jobs/life 0.344: life dense, work thin



Data: Amap aggregate indicators; no raw points published

③ Space syntax: configuration shapes movement

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▪ Xueyuanqiao 97.5%: the belt's mouth at the N4th Ring × Xueyuanlu

▪ Tsinghuayuan 93.5%: a 1909 siting still the structural core

▪ Dazhongsi 20.2%: syntactic periphery, vitality must come from function

▪ After stitching: Dazhongsi +70%, Beijing North +49%, Wudaokou +43%

5,720-node network, computed: low intelligibility is normal for organic fabric



④ Boundary correction: fixing the brief's rough boundary

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▪ The brief's PROV-KEY-003 rough boundary sits near Beijing North (39.944–39.950)

▪ Its own basis table names Dazhongsi Station (39.9653) — ~2 km away

▪ missing-data.md: precise polygons are missing

▪ Corrected by name semantics, assumption A-BOUNDARY-003, full recalculation on official data



Integrity first: a data mismatch must be disclosed, never quietly fixed



02

Spatial proposal: one line, three folds, two wings, seven nodes

Master plan · Key-area plans & sections · Character & height · Dazhongsi quadrants

⑤ Master plan (real fabric + concept overlay)

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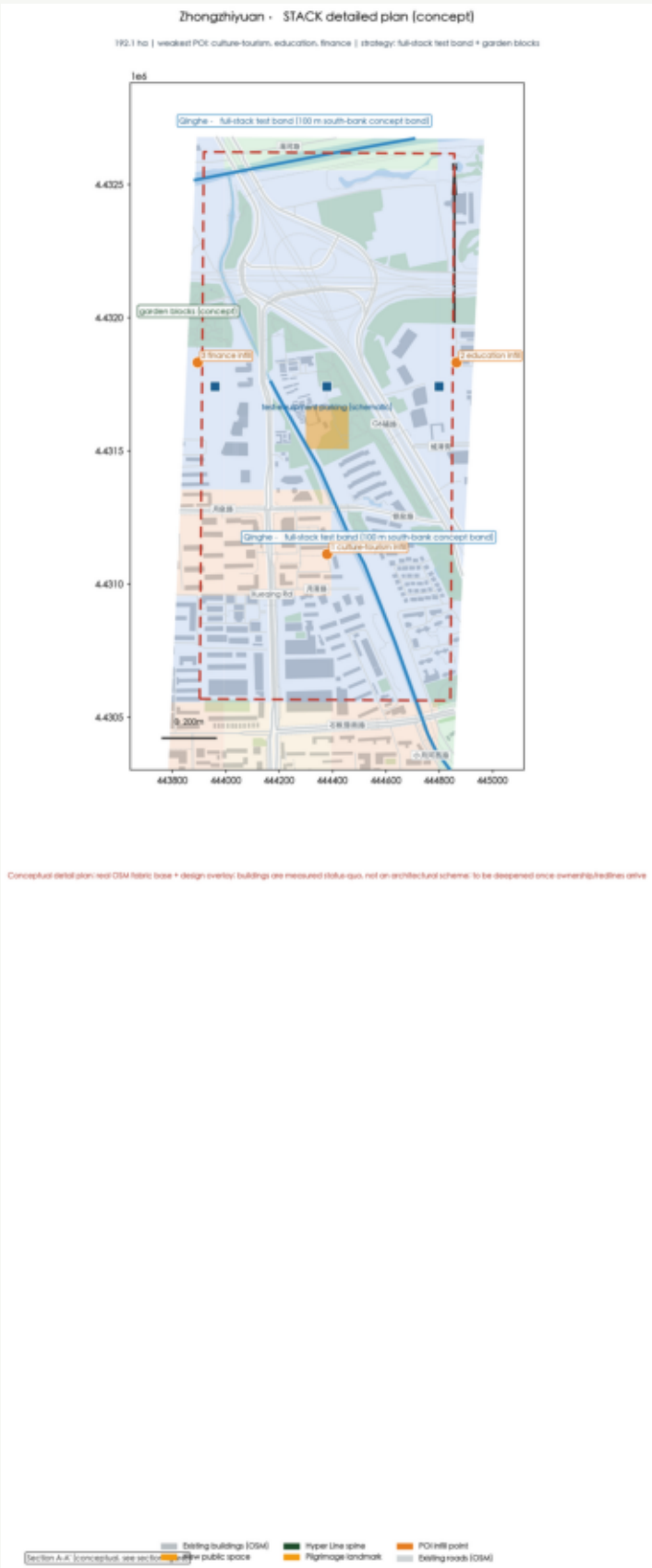


- Light grey = OSM building fabric; white-cased dark = real Jing-Zhang railway
- Dark green = Hyper Line spine; red dots = seven stitches; stars = three landmarks
- Three coloured frames = three gates: STACK / ORIGIN / FRONT
- The base is the real city; only the overlay is design

Base: OSM (© OpenStreetMap contributors)

⑥ Zhongzhiyuan · STACK (192.1 ha)

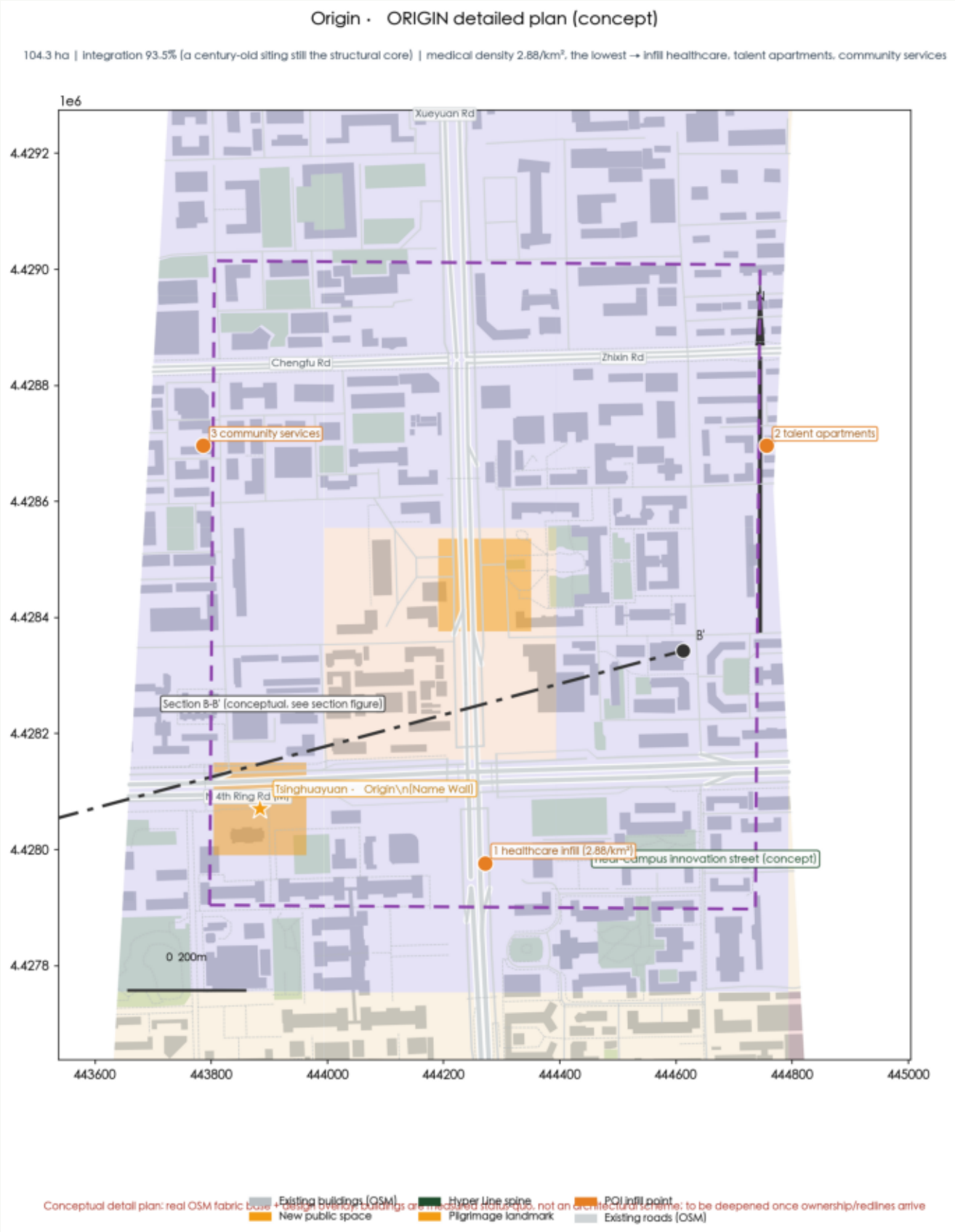
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- Qinghe full-stack test band: bounded tests in garden blocks
- Weakest: culture-tourism/education/finance → reasons to stay
- Section A-A': 3.2m footpath + 2.8m cycle band + test-equipment slots
- Long term (5–10y): garden-district renewal

⑦ Origin · ORIGIN (104.3 ha)

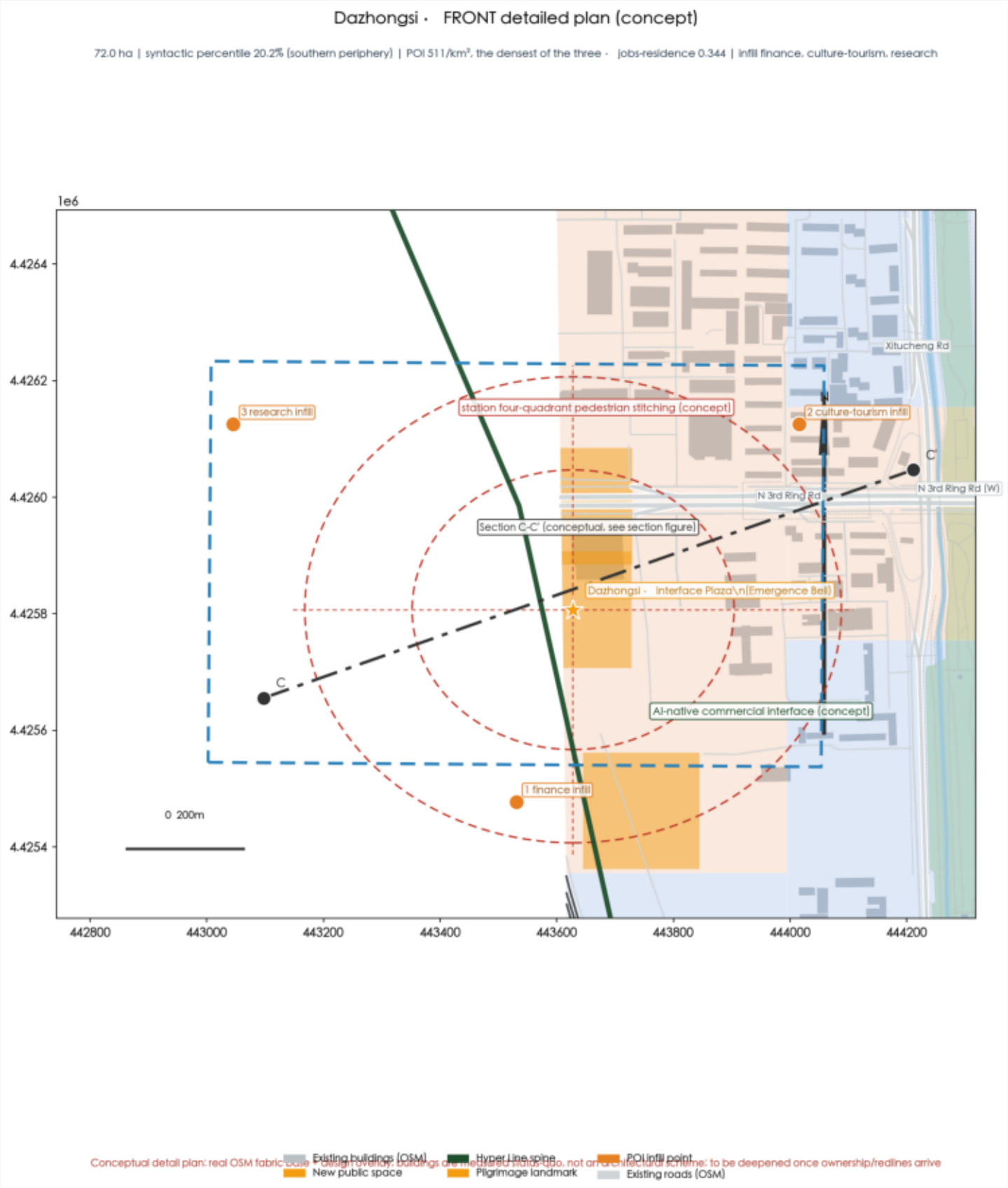
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- Tsinghuayuan · Origin Plaza + Name Wall: integration 93.5%
- Wudaokou · Phase Plaza: the λ gauge — the city's thermometer
- Medical 2.88/km², the lowest → infill healthcare/apartments/services
- Mid term (3–5y): near-campus innovation street

⑧ Dazhongsi · FRONT (72.0 ha)

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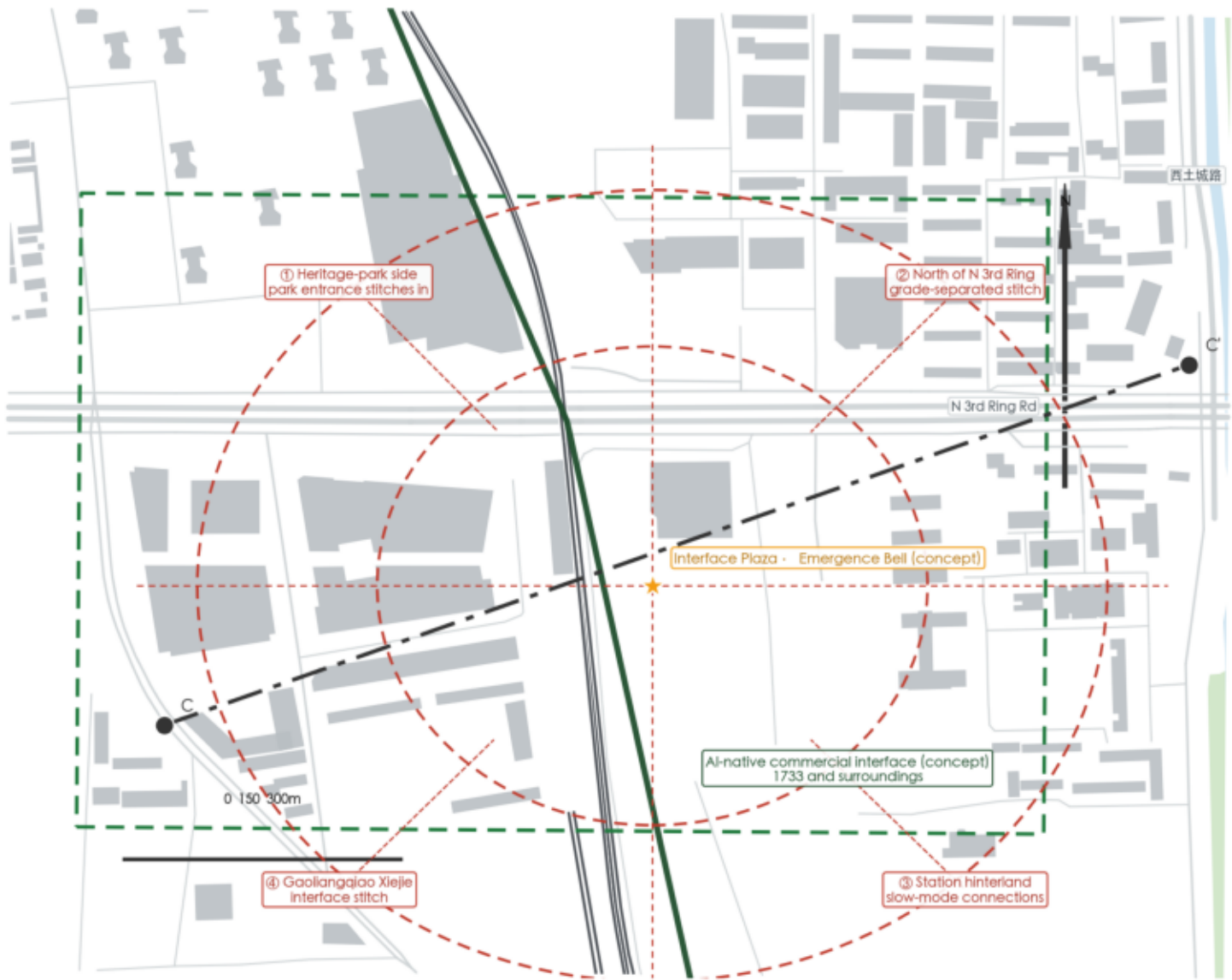


- Syntactic periphery 20.2%, yet the densest POIs at 511/km²
- Jobs/life 0.344: life-service-led → infill finance/culture-tourism/research
- Light intervention, heavy interface: stitch four quadrants: nights for classes & merchants
- Near term (1–3y): spine stitching + Dazhongsi interface

⑨ Dazhongsi four-quadrant stitching (enlarged)

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Fig C19 · Dazhongsi station-district four-quadrant stitching, enlarged (conceptual design)



Conceptual schematic: the stitch modes (grade-separation / slow-mode links) are conceptual suggestions, to be deepened once transport, municipal, and ownership data arrive; section C-C' shown in the detailed section figure.

Base: OSM (© OpenStreetMap contributors, ODbL)

- ① park entrance stitches in ② grade-separated stitch north of N3rd Ring
- ③ station-hinterland slow modes ④ Gaoliangqiao Xiejie interface
- Interface Plaza · Emergence Bell + the 1733 AI-native interface
- Section line C-C' matches the detailed plan

⑩ Street sections of the three areas (conceptual · dimension chain · plan index)

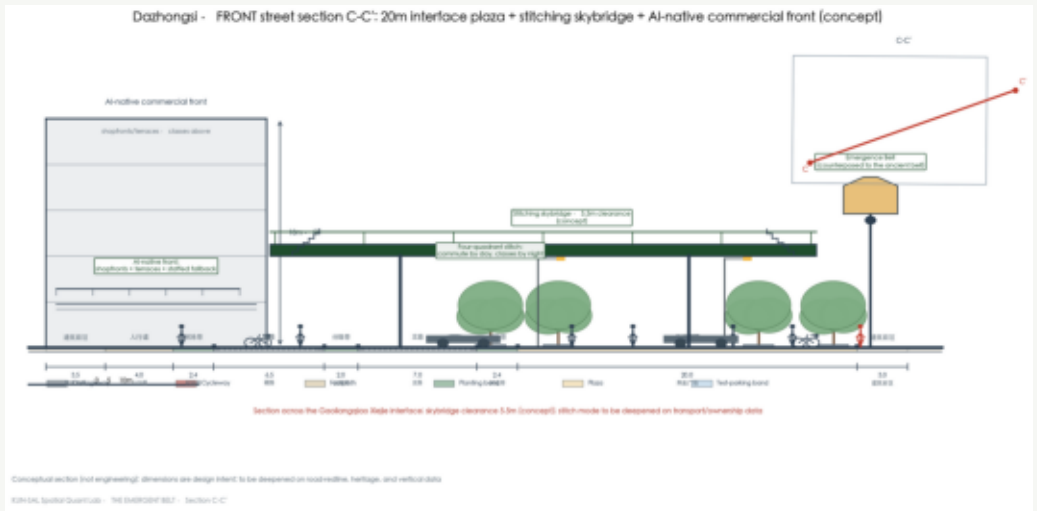
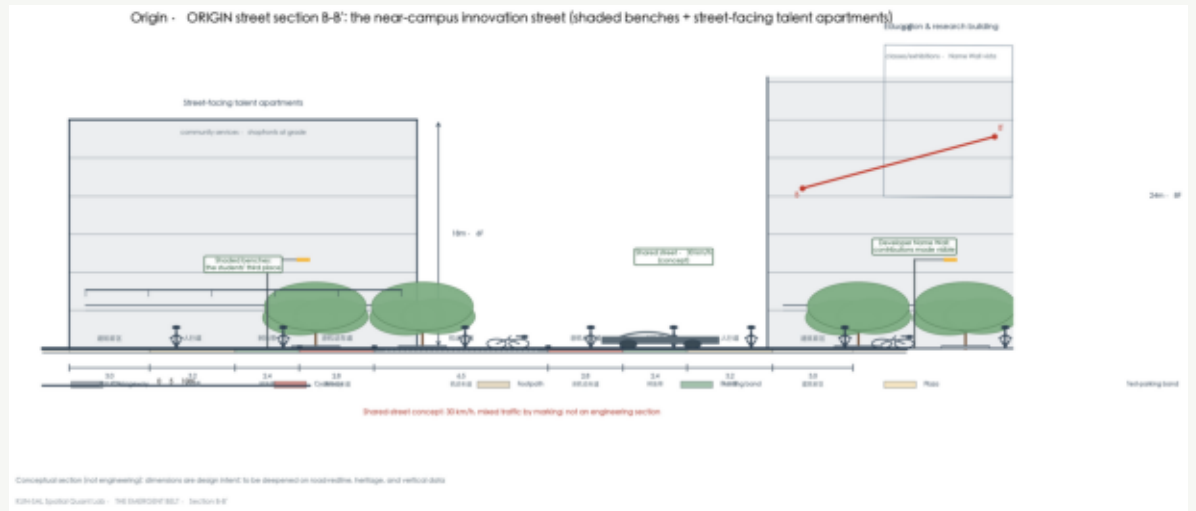
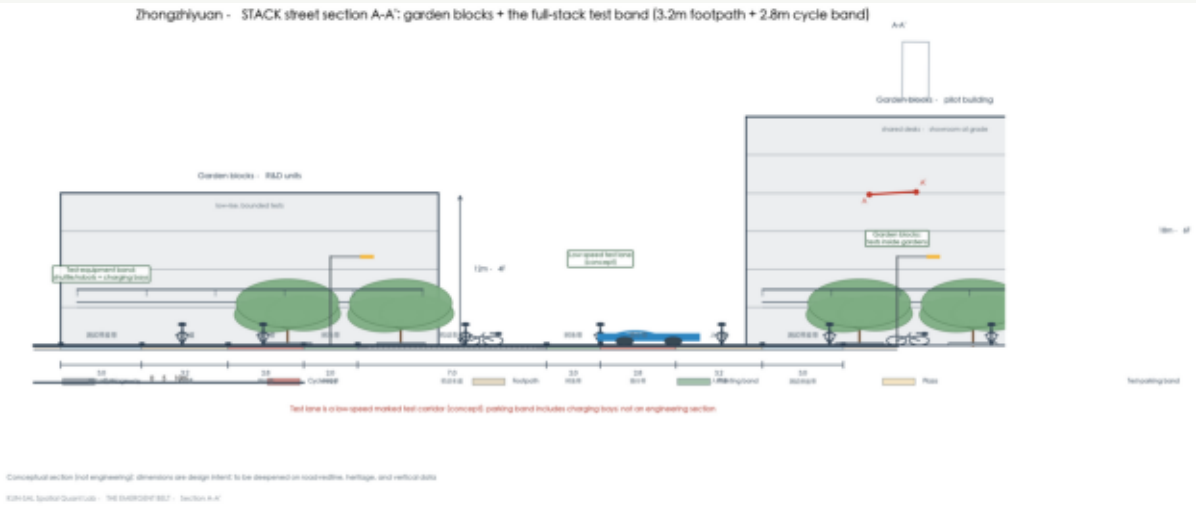
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▪ Each section: building-to-building full section + dimension chain + floors/heights

▪ Scale references: 1.8m people · bikes · cars · low-speed shuttle · trees & lamps

▪ Dazhongsi C-C': 20m interface plaza + stitching skybridge (5.5m) + Emergence Bell

▪ Conceptual sections (not engineering); dimensions pending redline & heritage data

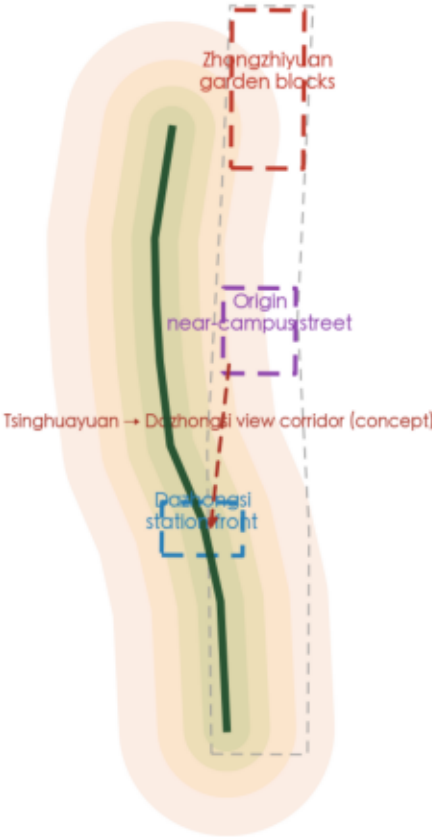


Conceptual sections: design intent; to be deepened on redline, heritage and vertical data

Character & conceptual height zoning

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Conceptual height zoning (schematic): bands step down toward the spine, low front / high back

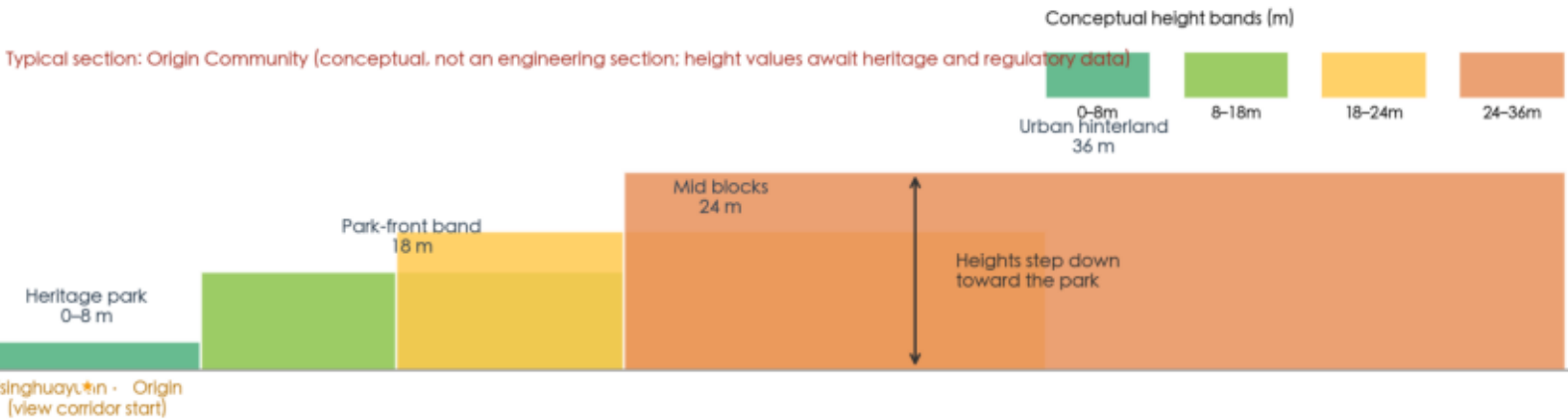


▪ Height bands step down along the spine: 0–8 / 18 / 24 / 36 m

▪ Three rules: garden blocks / near-campus street / station front

▪ Protective overlay (heritage + view corridors) + generative rules (low front, high back)

▪ Height values await heritage & regulatory data (not a regulatory drawing)



Schematic: height and character rules are design intent, to be rechecked once heritage, regulatory, and ownership data arrive (not a regulatory drawing)



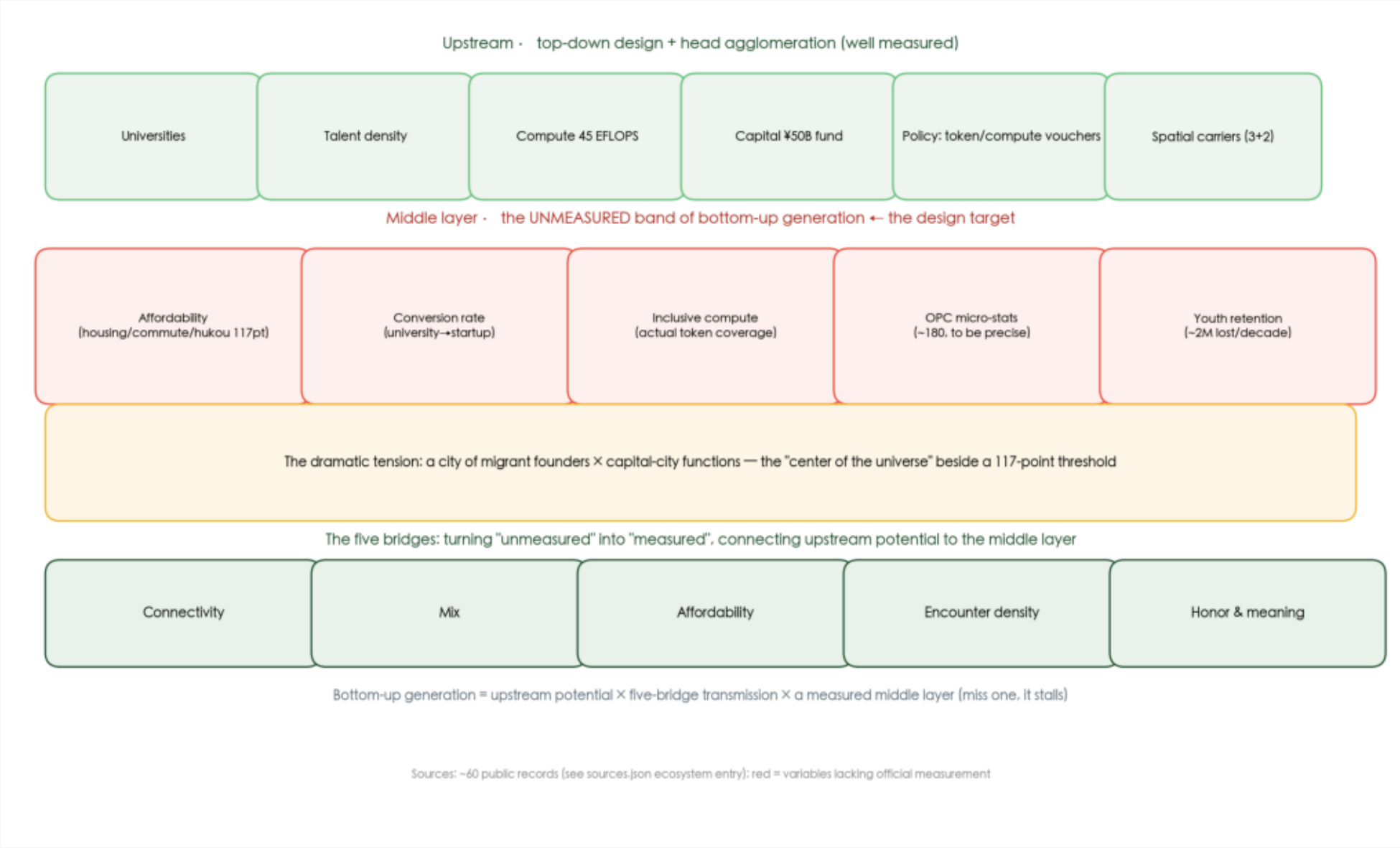
03

Scenarios & ecosystem: the AI ecology and pilgrimage

Ecosystem map & seven cases · Ten scenario cards, five personas · Three pilgrimage landmarks · Operations & timeline

AI innovation ecosystem map

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- Upstream complete: 1,000+ scientists, 13k developers, 180 OPCs, 32 chips
- Middle layer unmeasured: affordability/conversion/compute → annualized
- Seven cases, one transferable mechanism each (King's Cross / Kendall / Cambridge...)
- Five-party six-force conversion + the annual Phase-Forum report as interface

Operations & the ten-year timeline

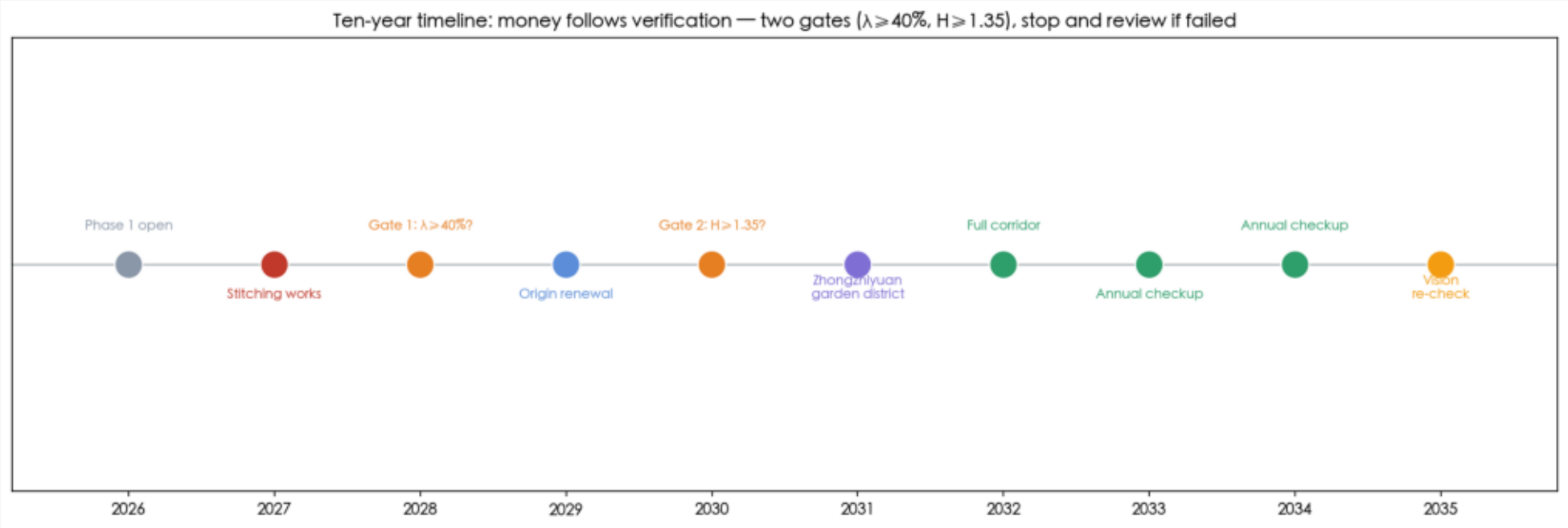
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▪ Annual: dev festival / Phase Forum (λ & H report) / open test seasons

▪ Attraction funnel: OPC contest → order plaza → desks → incubator → conversion

▪ Two acceptance gates: $\lambda \geq 40\%$, $H \geq 1.35$ — or stop and review

▪ Phases: 1–3 stitching+interface / 3–5 Origin / 5–10 Zhongzhiyuan



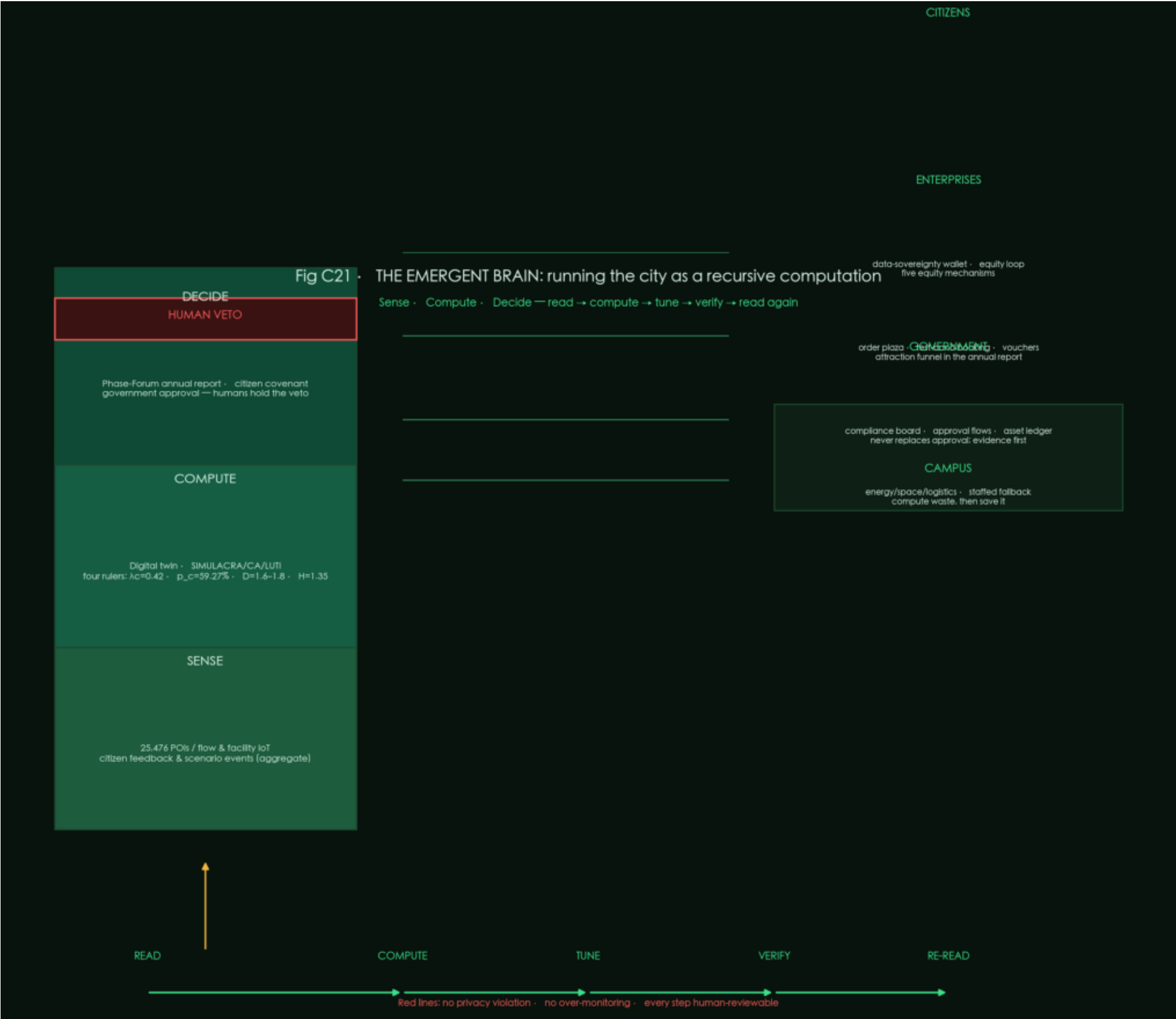
04

System & finance: the Emergent Brain and the investment ledger

Emergent Brain architecture · Phase Control Room · Investment & return · Risk & compliance

📄 The Emergent Brain: the city as a recursive program

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▪ Three layers: sense → compute → decide, with humans always holding the veto

▪ Four terminals: citizens / enterprises / government / campus

▪ The recursive loop is the annual check-up:
read→compute→tune→verify→re-read

▪ Red lines: no privacy violation, no over-monitoring, human-reviewable

Phase Control Room: the belt's annual check-up

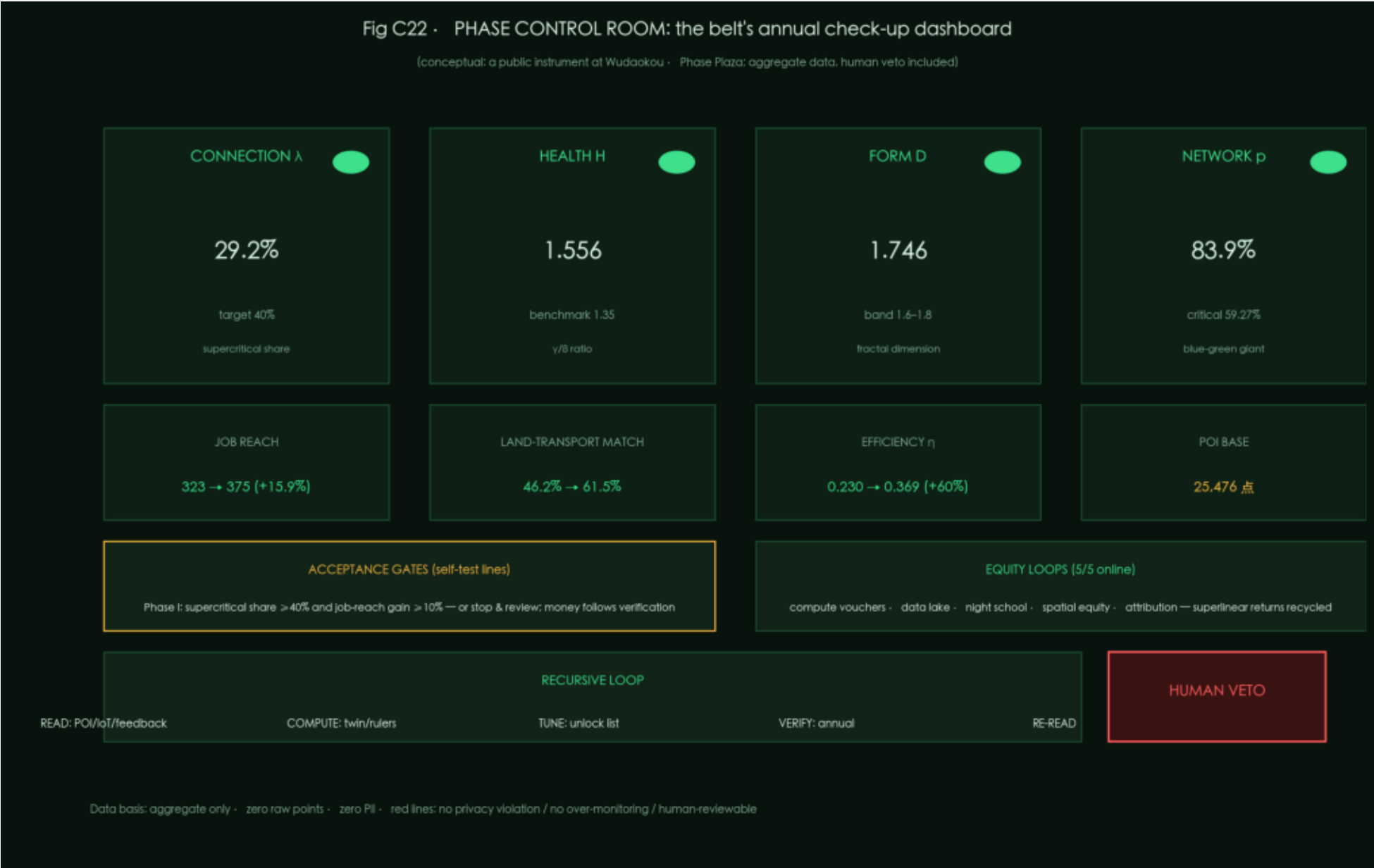
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▪ Four gauges: λ 29.2%→40% target / H 1.556 / D 1.746 / p 83.9%

▪ Live panels: job reach 323→375, match 46.2%→61.5%, η +60%

▪ Physically at Wudaokou · Phase Plaza: a public instrument, not surveillance

▪ Aggregate only, zero raw points, zero PII, human veto



Investment & return (conceptual magnitude)

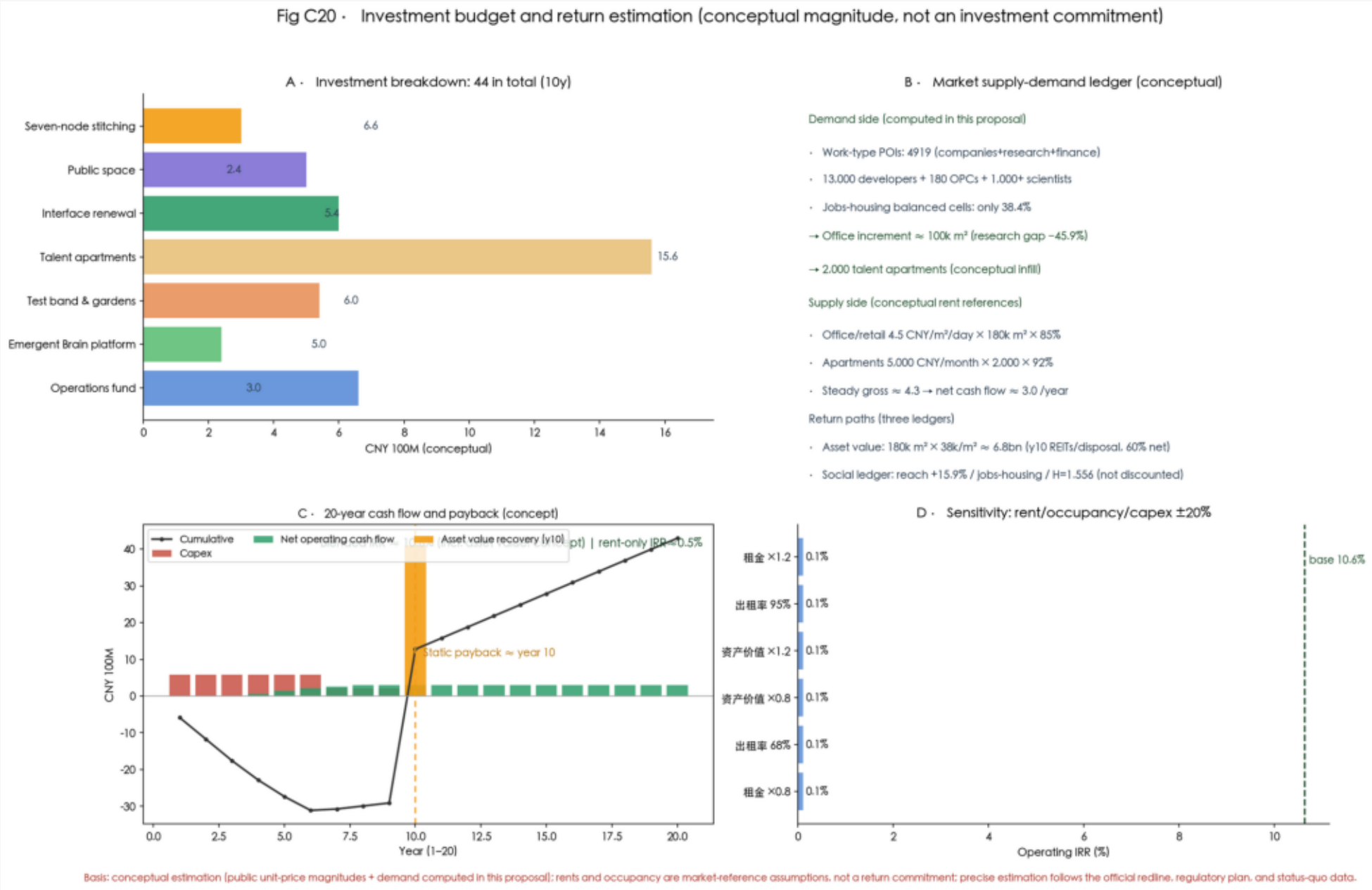
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CNY 4.40bn: stitching 0.66 / space 0.24 / interface 0.54 / apartments 1.56 / test band 0.60 / brain 0.50 / ops 0.30

Three ledgers: rent (IRR≈0.5% alone) + asset (y10 REITs/disposal, blended IRR≈10.6%) + social

Demand: 4,919 work POIs → 100k m² offices; 38.4% balance → 2,000 apartments

Public ledger ≈1.7bn seeks no financial return; market ledger ≈2.7bn runs rent+asset



Conceptual; not an investment commitment; rents & occupancy are replaceable assumptions



05

Conclusion: four sentences, each with numbers

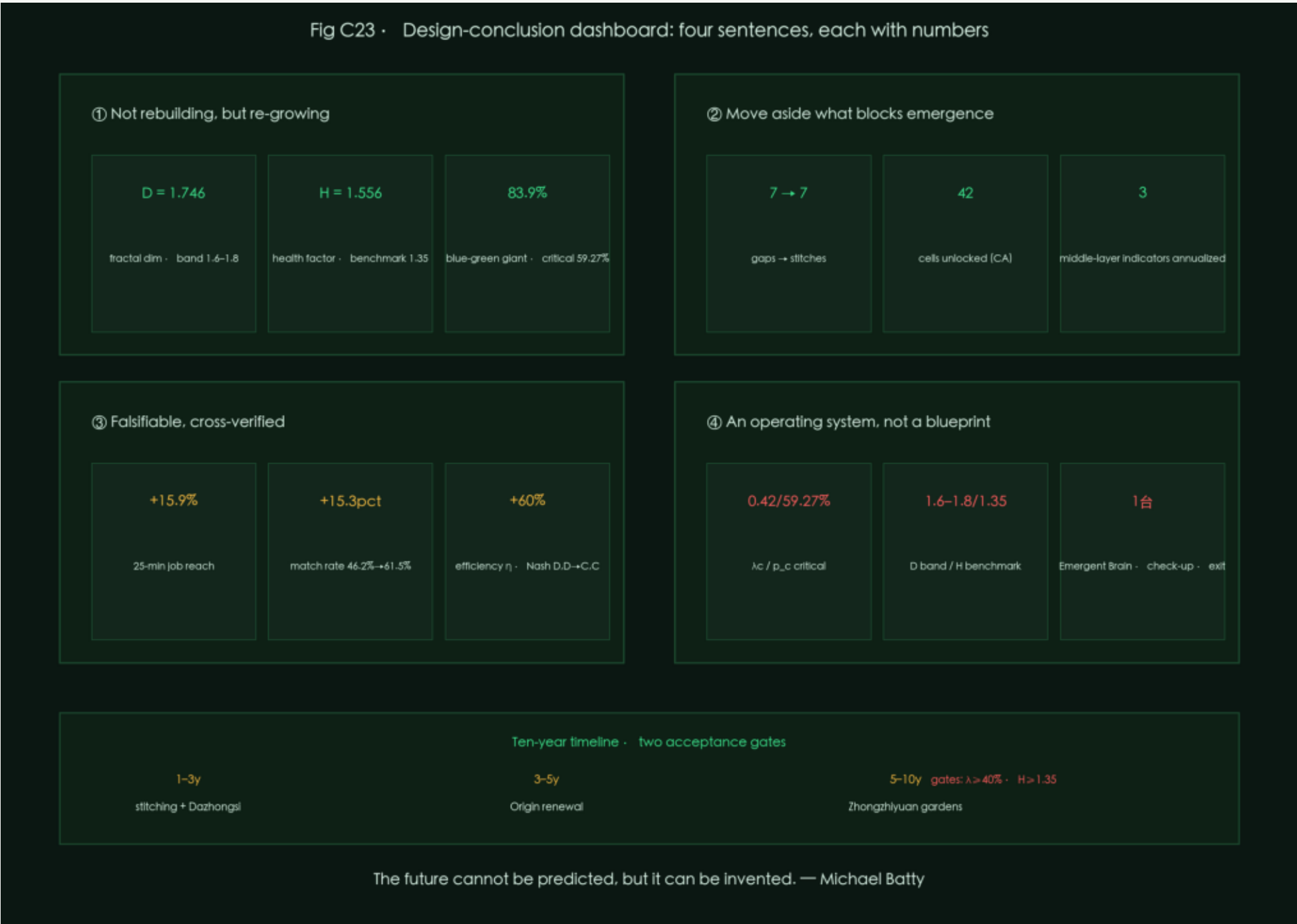
Conclusion dashboard · 1909 switch → 2026 belt

① Re-grow, not rebuild: D=1.746 / H=1.556 / giant 83.9%

② Move aside what blocks emergence

③ Falsifiable: jobs +15.9% / match +15.3pct / η +60%

④ An OS, not a blueprint: four rulers + brain + check-up + exit



The future cannot be predicted, but it can be invented.

— Michael Batty

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KUN-SAL Spatial Quant Lab · Xing Xinhua · [GitHub: Anshengdesign](#)

This proposal is an open co-creation suggestion; it does not replace formal planning and constitutes no government-reviewed conclusion; provisional boundaries are dashed only, statutory indicators pending.